On Simulating a 2-D Ideal MHD Fluid in an Initial Square Region

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1 Introduction

This project simulates the expansion of an ideal magnetohydrodynamic (MHD) plasma in two dimensions. The plasma is initially confined to a square region

$$\Sigma = [25, 75]^2$$

within a larger domain

$$\Omega = [0, 100]^2.$$

At time $t = 0$, the plasma is released into vacuum by removing the boundaries of Σ, and it expands into Ω subject to the ideal MHD equations.

We track the evolution of key quantities: density ρ, pressure p, velocity $\vec{V}$, and magnetic field $\vec{B}$. The simulation allows visualization of plasma dynamics and interactions with $\partial\Omega$, the boundary of the domain.

2 Governing Equations

The system evolves according to the ideal MHD equations [1]:

$$\frac{d\rho}{dt} + \rho\nabla\cdot\vec{V} = 0 \qquad \text{(Mass conservation)}$$

$$\rho\frac{d\vec{V}}{dt} + \nabla p - \vec{j}\times\vec{B} = 0 \qquad \text{(Momentum conservation)}$$

$$\vec{E} + \vec{V}\times\vec{B} = 0 \qquad \text{(Ideal Ohm's Law)}$$

$$\frac{d}{dt}\left(\frac{p}{\rho^\Gamma}\right) = 0 \qquad \text{(Adiabatic law)}$$

where $\Gamma = \frac{5}{3}$ for an ideal monatomic gas, and $\vec{j} = \frac{1}{\mu_0}\nabla\times\vec{B}$.

3 Numerical Method

We implemented a finite-difference explicit solver in Python using central differencing for spatial derivatives and forward Euler time-stepping. The domain was discretized into a 100×100 grid, with plasma initialized uniformly within Σ, and magnetic field $\vec{B} = B_0\hat{y}$. The outer boundary $\partial\Omega$ used open (zero-gradient) conditions.

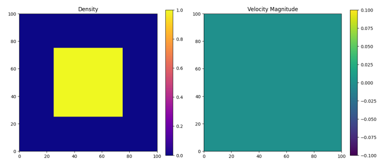
To ensure stability, we enforced:

$$\rho \geq \rho_{\text{floor}} = 10^{-3}, \quad p \geq 10^{-5}, \quad |\vec{V}| \leq 5$$

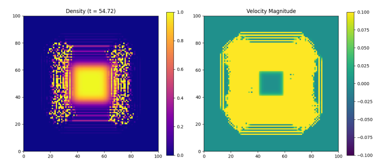
and computed the time step dynamically using the CFL condition.

4 Results

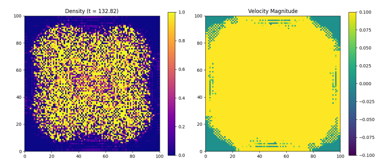
The following figures show the plasma state at three stages of the simulation: initial, mid, and late time.



(a) Initial state



(b) Mid expansion



(c) Late expansion

Figure 1: Density (left) and velocity magnitude (right) at selected times.

The plasma expands outward symmetrically from Σ, accelerating under its own pressure gradient. Velocity reaches a maximum near the edges of the expansion front. Magnetic tension opposes expansion perpendicular to $\vec{B}$, leading to mild anisotropy over time.

5 Conclusion

This project demonstrates that even a simple 2-D ideal MHD model captures rich plasma dynamics such as anisotropic expansion, density gradients, and energy transport. The simulation was numerically stabilized via pressure floors, time-step control, and clamping to avoid overflows, and produced realistic results consistent with physical expectations.

References

[1] Richard Fitzpatrick. *Plasma Physics: An Introduction*. Second. Referenced Chapter 8: Magnetohydrodynamic Fluids, equations 8.1–8.4. CRC Press, 2022. ISBN: 978-1-032-20251-8.